

DRUBO DEV PRAMANIK

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🌐 LinkedIn | 🐙 Github | 🌐 Website

RESEARCH INTERESTS

- Federated Learning
- Deep Learning
- Image Segmentation
- Compression and Quantization

EDUCATION

Rajshahi University of Engineering & Technology (RUET) Rajshahi, Bangladesh
B.Sc. in Computer Science & Engineering 2022 – 2026
CGPA: **3.72** out of 4.00

New Government Degree College Rajshahi, Bangladesh
HSC in Science | GPA: **5.00** out of 5.00 2018 – 2020

BIAM laboratory School & College Bogura, Bangladesh
SSC in Science | GPA: **5.00** out of 5.00 2015 – 2018

THESIS

Undergraduate Thesis August 2025 – August 2026

Thesis Title: Communication-Efficient Federated Learning for Brain Tumor Segmentation Using Noise-Aware Adaptive Compression.

Developed **CAN-FL**, a communication-efficient federated learning framework for privacy-preserving brain tumor segmentation using adaptive 4/6/8-bit compression and noise-aware aggregation, evaluated on the FeTS2022 MRI dataset.

RELEVANT COURSEWORK

- Data Structures and Algorithms
- Object-Oriented Programming
- Database Management Systems
- Operating Systems
- Computer Networks
- Artificial Intelligence

PUBLICATIONS

CAN-FL: Communication-Efficient Federated Brain Tumor Segmentation via Noise-Aware Adaptive Compression

Drubo Dev Pramanik, Md. Shahid Uz Zaman

Accepted at IEEE SPICSCON 2026

PROFESSIONAL EXPERIENCE

Codingal September 2026 – Present
Coding Instructor [Link](#)

- Teach students up to Grade 11–12 in Python, C, C++, Java, AI, and other programming fundamentals.
- Guide students through hands-on projects while developing problem-solving and computational thinking skills.
- Mentor students for coding contests and support their progress through personalized feedback.

Netro Systems June 2025 – December 2025
Industrial Attachment Intern

- Worked with backend development workflows using PHP/Laravel and MySQL database management.
- Assisted in software testing, debugging, and technical documentation to improve application reliability.
- Collaborated with development teams following structured software development practices and version-controlled workflows.

SELECTED PROJECTS

CoffeeMate AI — Personalized Coffee Shop Assistant

🐙 [GitHub](#)

Python, Google ADK, Gemini, RAG, Google Cloud Run

- Developed a customer-facing AI coffee assistant using the Google Agent Development Kit (ADK) and Gemini to provide personalized coffee recommendations through natural-language conversations.
- Implemented Retrieval-Augmented Generation (RAG) over a coffee-shop knowledge base to retrieve menu, pricing, ingredient, dietary, and policy information and reduce hallucinations.
- Deployed the application on Google Cloud Run with a responsive web interface, enabling customers to receive recommendations based on conversational preferences such as strength, sweetness, milk, temperature, and caffeine.

Undergraduate Student CGPA Prediction

 [GitHub](#)

Python, Flask, Pandas, NumPy, Machine Learning, HTML, CSS

- Developed a Flask-based machine-learning web application for predicting university students' CGPA from structured academic data.
- Performed data preprocessing and evaluated multiple predictive models using Python, Pandas, NumPy, and machine-learning techniques.

Real-Time Face and Eye Detection Web Application

 [GitHub](#)

Python, Flask, OpenCV

- Developed a Flask-based computer-vision application capable of streaming live webcam video through a web interface.
- Implemented real-time face and eye detection using OpenCV Haar cascade classifiers.

TECHNICAL SKILLS

Programming Languages	C, C++, Python, Java, PHP, 8086 Assembly
Machine Learning & Data	PyTorch, scikit-learn, Pandas, NumPy, MONAI, CUDA, OpenCV
Web Development	Flask, Laravel, HTML, CSS, JavaScript
Databases	MySQL, SQL, Relational Database Design, DBMS

LEADERSHIP & ACTIVITIES

Senior Executive Member , Anuranan Cultural Club, RUET	2024 – 2026
Senior Executive Member , Band Music Society of RUET (BMSR)	2025 – 2026
Lead Vocalist , Chakravuyh Rock Band	2026
Volunteer , RUET Tech Fest Organizing Committee	2023

AWARDS & ACHIEVEMENTS

- Secured **1st position in the Department of Computer Science & Engineering** in the 7th semester with a **CGPA of 3.99/4.00**.
- Received a **Certificate of Participation in Career Compass 360°**, organized by Rajshahi University Business Club. 2025
- Received **Government Scholarship** for securing outstanding results in the HSC examination. 2020
- Received **Government Scholarship** for securing outstanding results in the SSC examination. 2018

REFERENCES

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